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# SeDA: Secure and Efficient DNN Accelerators with Hardware/Software Synergy

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- Introduction
- Motivation
- Solution
- Evaluation

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# Secure DNN Accelerators is important

(1) Autonomous vehicles are susceptible to cyberattacks that can disrupt critical functions and pose serious safety risks





(3) Healthcare cybersecurity threats are escalating, endangering patient data, medical devices, and operations

(2) Maliciously hacked AI robots may harm human interests

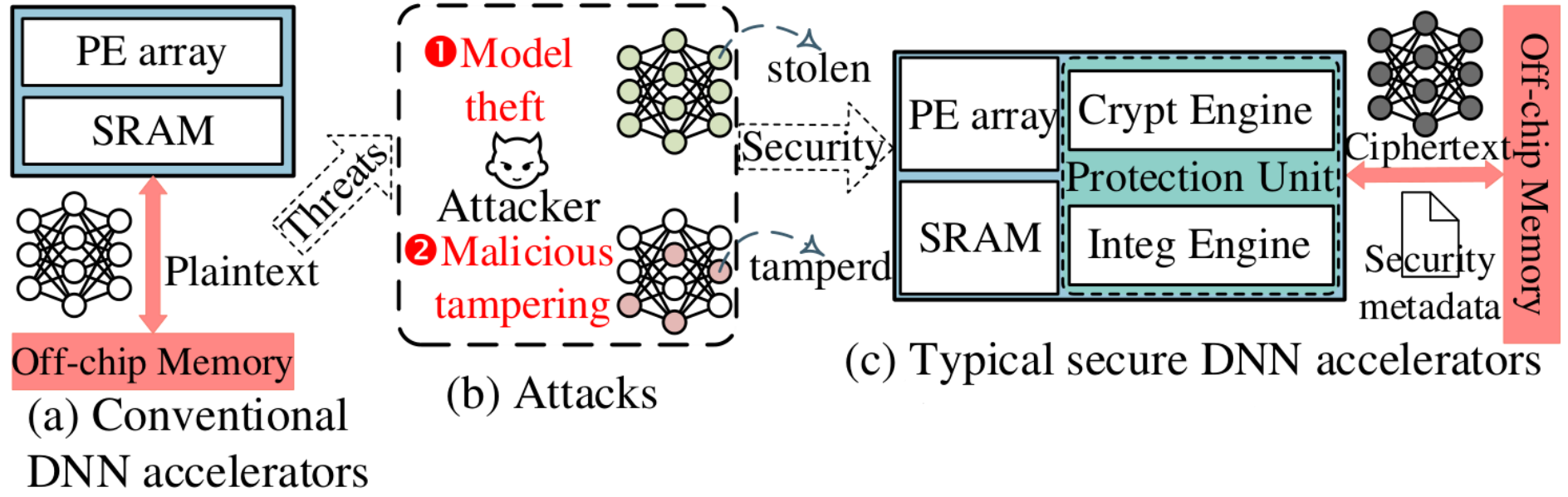




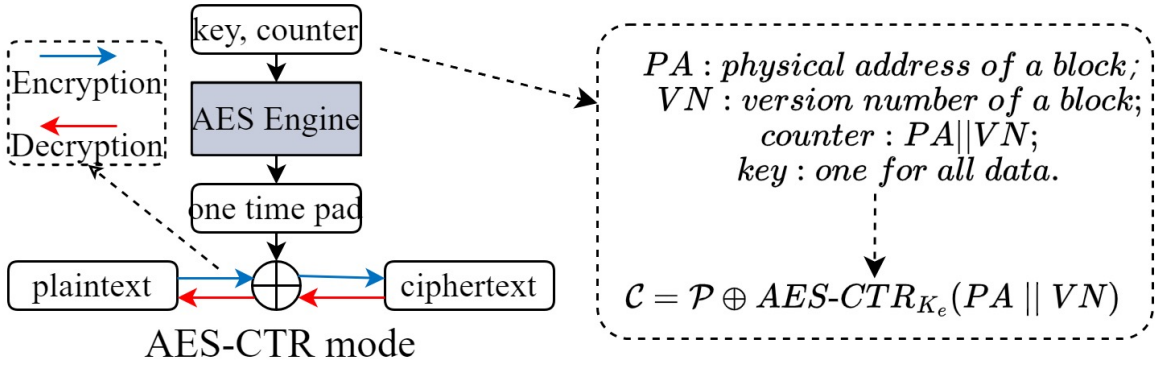
(4) Malicious data alterations by attackers undermine financial market fairness and investor trust

# Memory Protection Mechanism

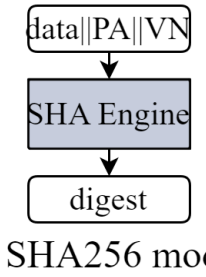
- **Goal:** augment traditional unsecure DNN accelerators with the capability to cryptographic authentication



# Authentication En/Decryption



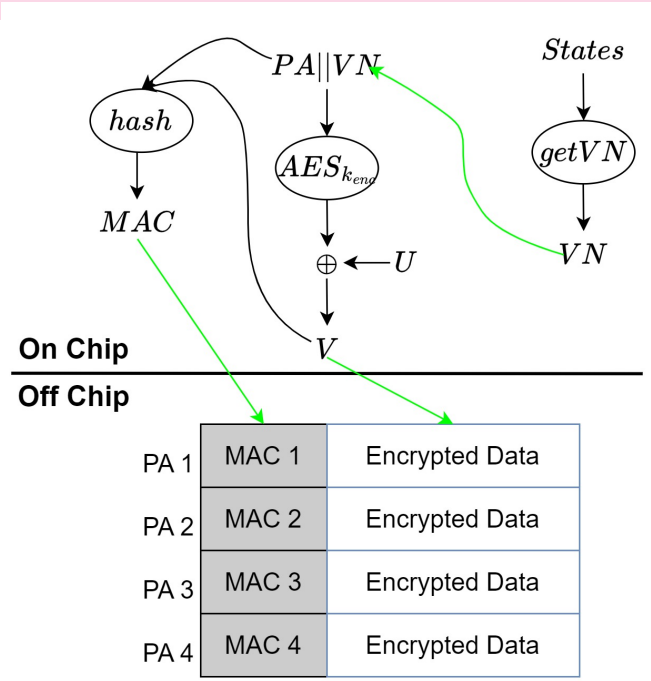

Confidentiality Protection




$MAC$ : *message authentication code*

$MAC = SHA256(data \,||\, PA \,||\, VN)$

Integrity Verification




- Confidentiality protection (CP) and Integrity Verification (IV) both involve the current data block's PA and VN.
- Introduce memory overhead for MACs.

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# Hardware Overhead of Security

- Secure DNN accelerators can **model** cryptographic and authentication operations as an additional constraint on off-chip DRAM bandwidth [1]
- **Effective memory bandwidth** = Min (original memory bandwidth, memory protection bandwidth)

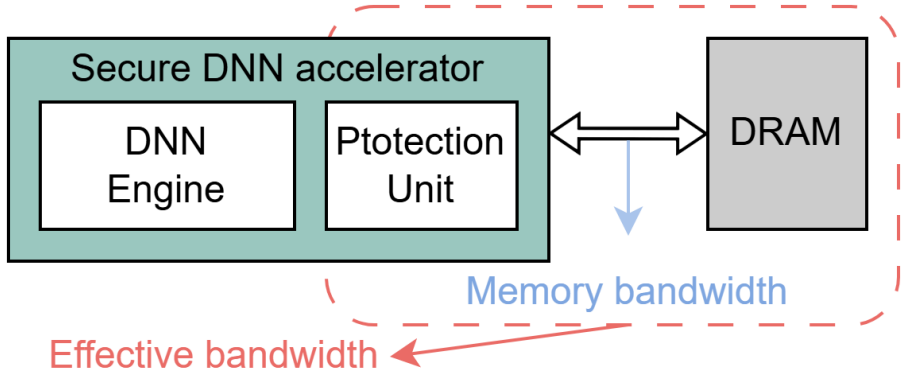
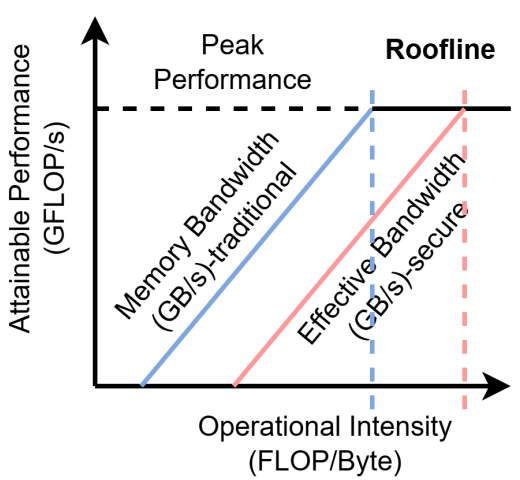
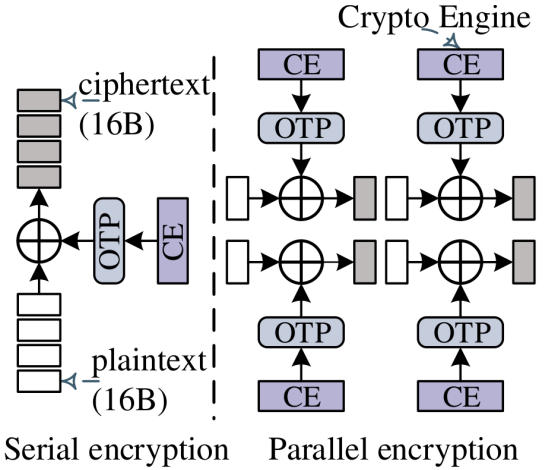



Diagram of effective memory bandwidth

- Analysis via the roofline model reveals that the **bandwidth of security protection unit** could constrain the performance of secure DNN accelerators
- Parallel hardware encryption incurs overhead to meet bandwidth needs of accelerators




Roofline model analysis



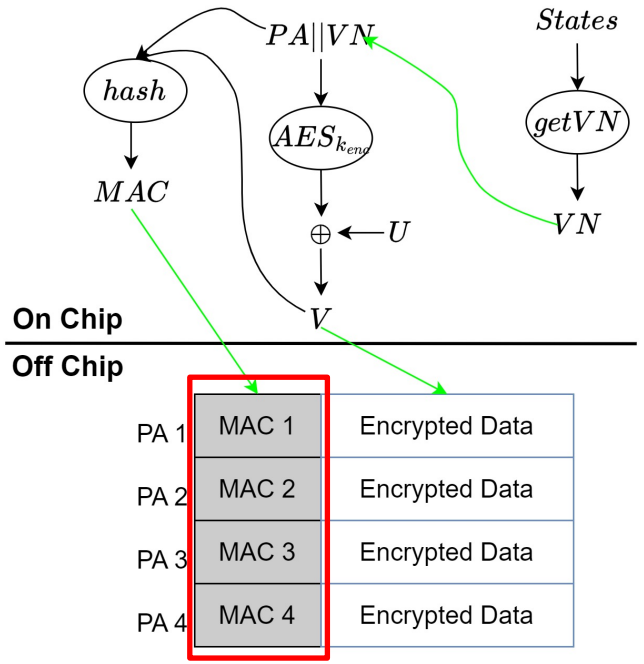

Serial & parallel hardware implement

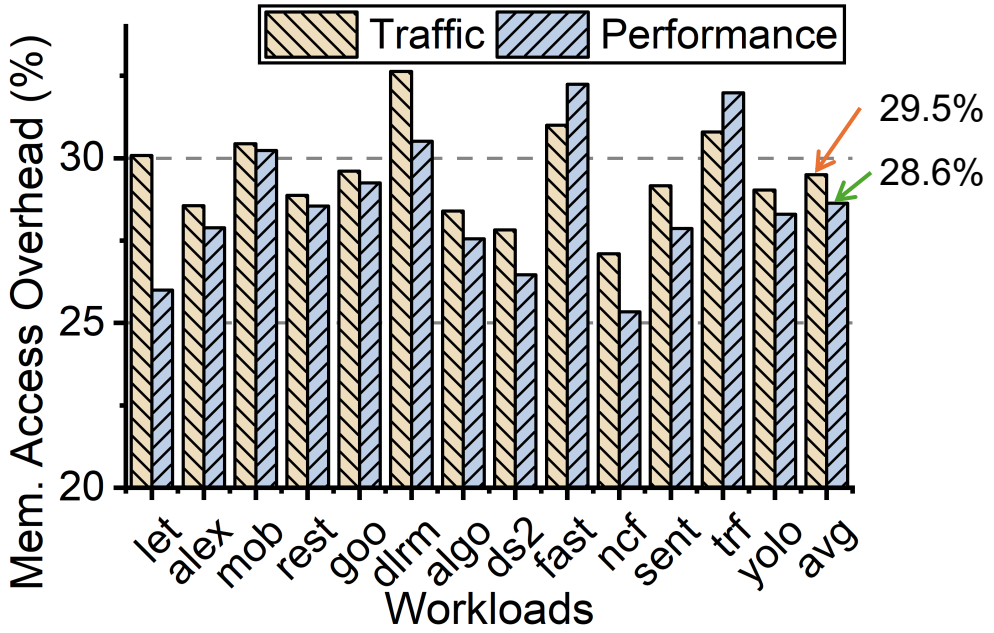
[1] SecureLoop: Design Space Exploration of Secure DNN Accelerators

# Extra Off-chip Memory Access

Costly off-chip memory access overhead due to **security metadata**:

- Memory traffic increases 29.5% on average
- Performance decreases 28.6% on average




Memory access overhead of security metadata

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# Key Insight 1: Multi-OTP Generation

- **High hardware resources** for confidentiality protection to provide high en/decryption bandwidth
- Shared OTP incurs **Single-Element Collision Attack (SECA)**

**Attack of SECA**

**Input** : OTP: one time pad for a data block ($blk$);
$most_value_p$: most used plaintext in $blk$.

**Output** : $value_p$: all plaintext of $blk$

1 $most_value_c \leftarrow$ CALCFREQVALUE($blk$)
2 OTP$\leftarrow most_value_p \oplus most_value_c$
3 **for** *each encrypted element* $value_c$ *of* $blk$ **do**
4 $\quad value_p \leftarrow value_c \oplus$ OTP

Minimal hardware overhead, supporting multiple distinct OTPs generation

# Key Insight 2: Cross-layer Authentication

- **High hardware resources** for confidentiality protection to provide high en/decryption bandwidth
- Shared OTP incurs **Single-Element Collision Attack (SECA)**
- Cross-layer different tiling patterns incurs error of integrity check

**Attack of RePA**

**Input :** $\mathrm{MAC}_i$: the MAC of a data block in one layer.

**Output :** $plaintext_e$: error plaintext.

$\mathrm{SUM_MAC} \leftarrow \sum_{i=1}^{n} \oplus \mathrm{MAC}_i$

$\textsc{ShuffleOrder}(\mathrm{MACs})$

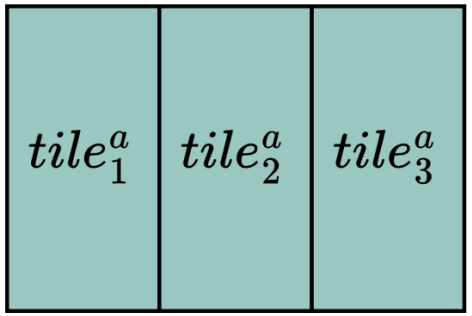
$\mathrm{SUM_MAC_shuffle} \leftarrow \sum_{i=1}^{n} \oplus \mathrm{MAC}_i'$

**if** $True = \textsc{VerifyInteg}(SUM_MAC, SUM_MAC_suffle)$ **then**

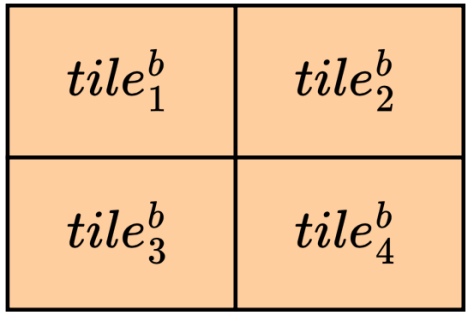
  **for** *each encrypted data block blk of one layer* **do**

    $plaintext_e \leftarrow \textsc{Decrypt}(blk)$

Different tiling patterns for the same matrix



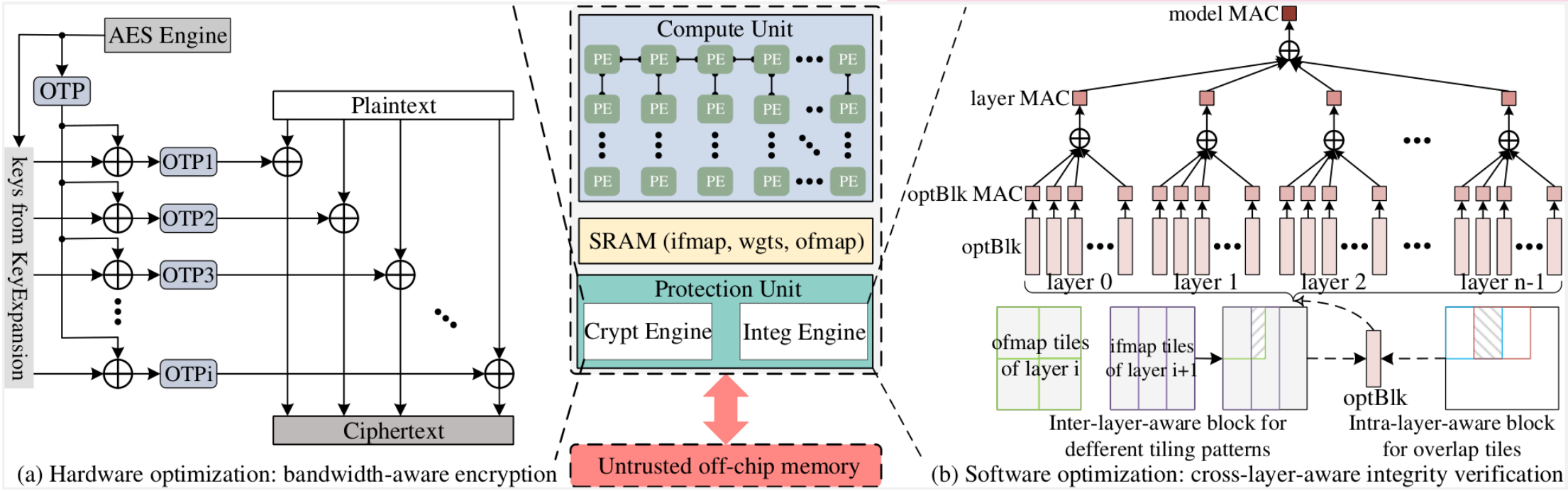

(1) Tiling pattern: $a$




(2) Tiling pattern: $b$

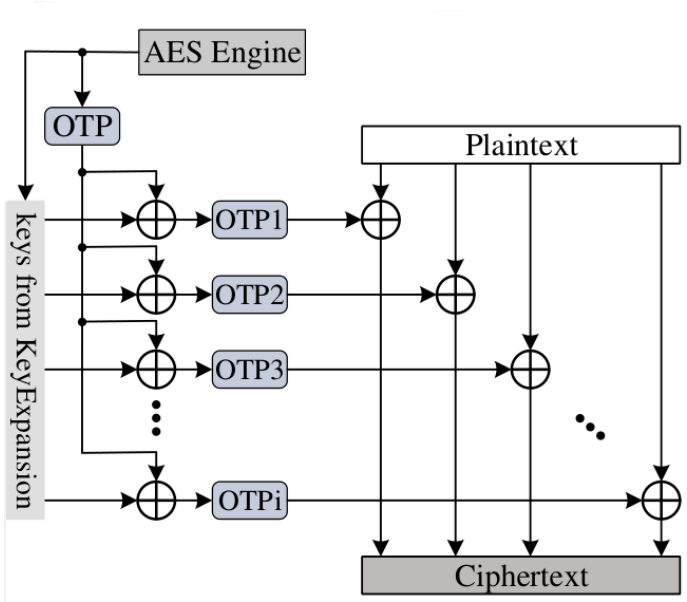
Cross-layer authentication considering different tiling patterns

# Overview of This Work




- (a) The Crypt Engine optimizes hardware by XORing keys from the AES Engine's KeyExpansion module with OPT, creating bandwidth-sensitive encryption granularity and reducing hardware overhead compared to using multiple AES Engines.
- (b) The Integ Engine optimizes software by analyzing overlapping tiles within a layer and patterns across layers to determine the optimal block size, optBlk, for integrity verification.

# Bandwidth-aware Encryption




**Defense of SECA**

**Input** : PA, VN: physical address and version number of $blk$.

**Output** : $\text{OTP}_i$: generate multiple OTPs for $blk$

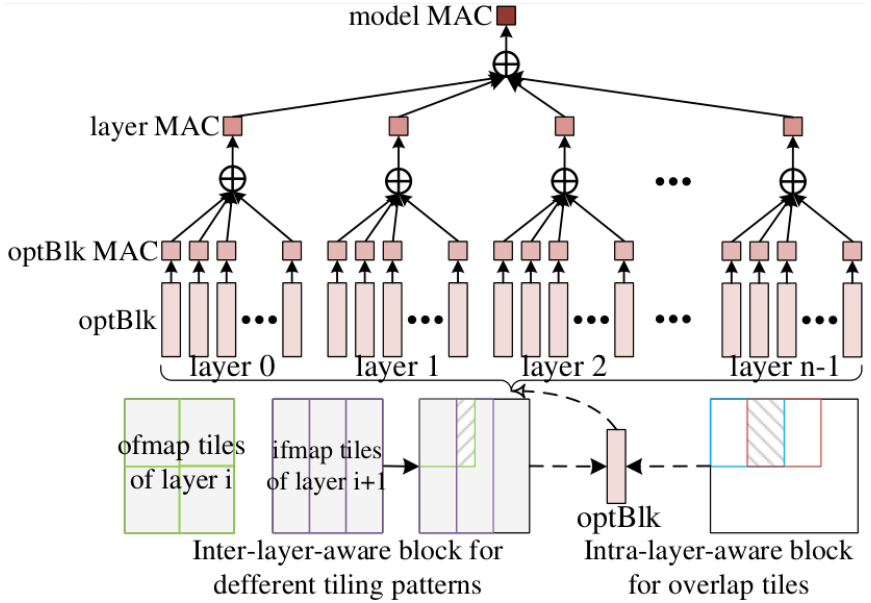
5 $\text{OTP} \longleftarrow \text{AES-CTR}_{K_e}(\text{PA} \parallel \text{VN})$

6 **for** *each 128-bit* $key_i$ *in keyExpansion of AES-CTR* **do**

7 $\quad \text{OTP}_i \longleftarrow (\text{OTP} \oplus key_i)$

- Leverage the inherent features of AES-CTR encryption mode
- Use just a single AES engine and a minimal number of XOR logic gates to generate multiple distinct OTPs at KeyExpansion
- Provide high effective bandwidth while avoid SECA

# Multi-level Integrity Verification




**Defense of RePA**

**Input** : $layer_{id}$, $fmap_{idx}$, $blk_{idx}$: layer number, feature map index and block index of $layer_{id}$.

**Output** : Secure layer MAC.

**for** *each encrypted data block* $blk$ *of* $layer_{id}$ **do**

  $\text{MAC}_i \longleftarrow \text{HASH}_{K_h}(blk \parallel \text{PA} \parallel \text{VN} \parallel layer_{id} \parallel fmap_{idx} \parallel blk_{idx})$

- Explore optimal block (optBlk) of authentication using Secureloop [1], considering diverse inter-layer tiling patterns
- Calculate message authentication code of each optBlk (optBlk MAC)
- Aggregate layer MAC (model MAC) by XORing all optBlk MACs (layer MACs)
- Minimize the off-chip memory access to only layer MACs

[1] SecureLoop: Design Space Exploration of Secure DNN Accelerators

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# Evaluation Methodology

- Simulators:
  - SCALE-Sim2 [1], Ramulator2 [2]
- Workloads:
  - Lenet, Alexnet, Mobilenet, ResNet18, GoogleNet, DLRM, AlphaGoZero, DeepSpeech2, FasterRCNN, NCF_recommendation, Sentimental_seqCNN, Transformer_fwd, Yolo_tiny
- Accelerator configuration:
  - Server NPU (Google TPU v1), Edge NPU (Samsung Exynos 990)

| Metrics | Server (Google TPU v1) | Edge (Samsung Exynos 990) |
|---|---|---|
| **PE** | 256 x 256 in systolic array | 32 x 32 in systolic array |
| **Bandwidth** | 20 GB/s with 4 channels | 10 GB/s with 4 channels |
| **Frequency** | 1 GHz | 2.75 GHz |
| **SRAM** | 24 MB | 480 KB |
| **Precision** | 1 B for per element | 1 B for per element |

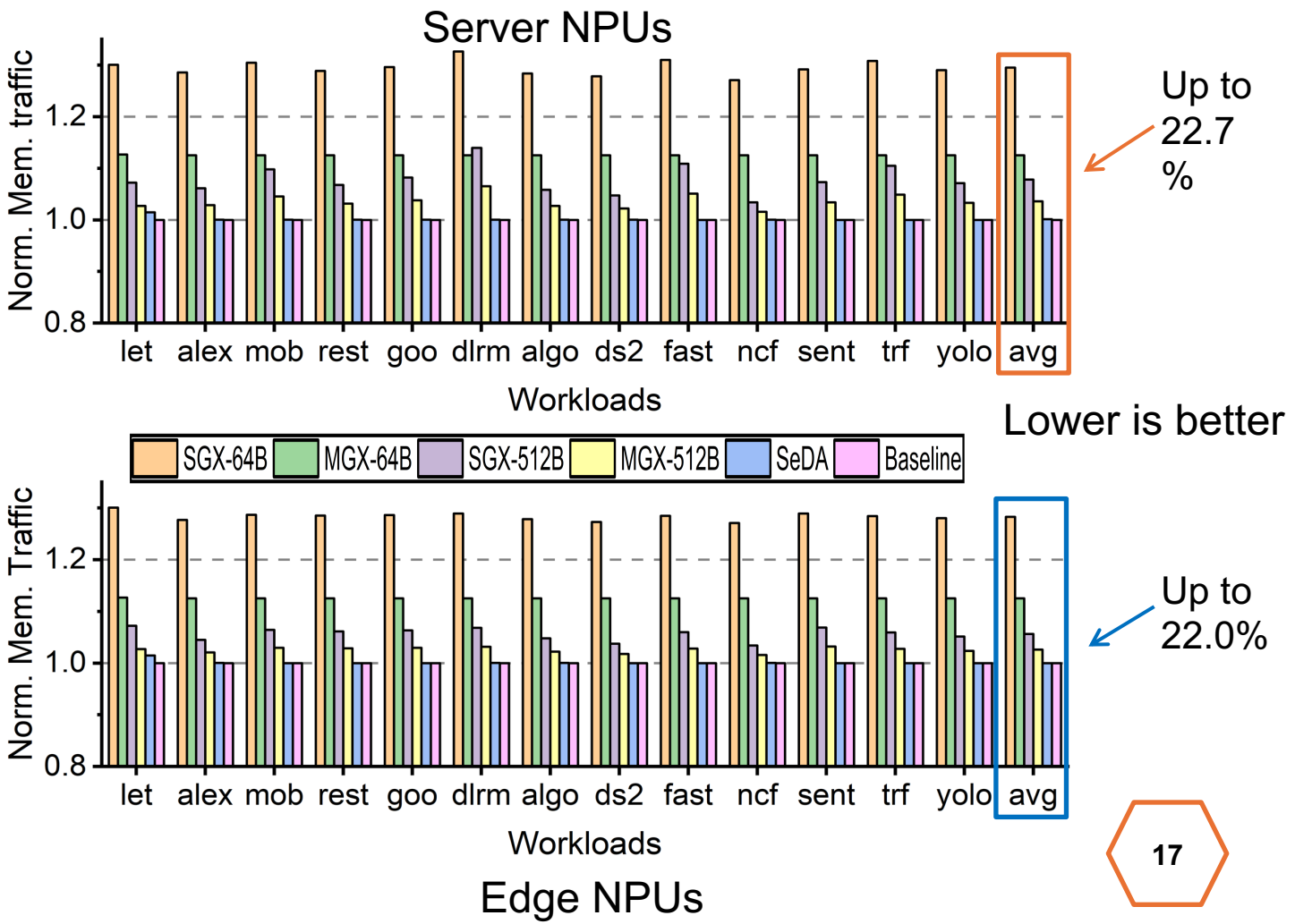
| Authentication en/decryption | Mode | Cycle |
|---|---|---|
| **AES-GCM** | parallel | 19 cycles [3] |

[1] A systematic methodology for characterizing scalability of DNN accelerators using SCALE-sim
[2] Ramulator 2.0: A modern, modular, and extensible DRAM simulator.
[3] Energy-efficient protocols and hardware architectures for transport layer security

# Traffic Comparison

- Traffic comparison with five protection schemes:
  - SGX-64B, SGX-512B, MGX-64B, MGX-512B and unsecure baseline
  - 64B and 512B indicate different security granularities

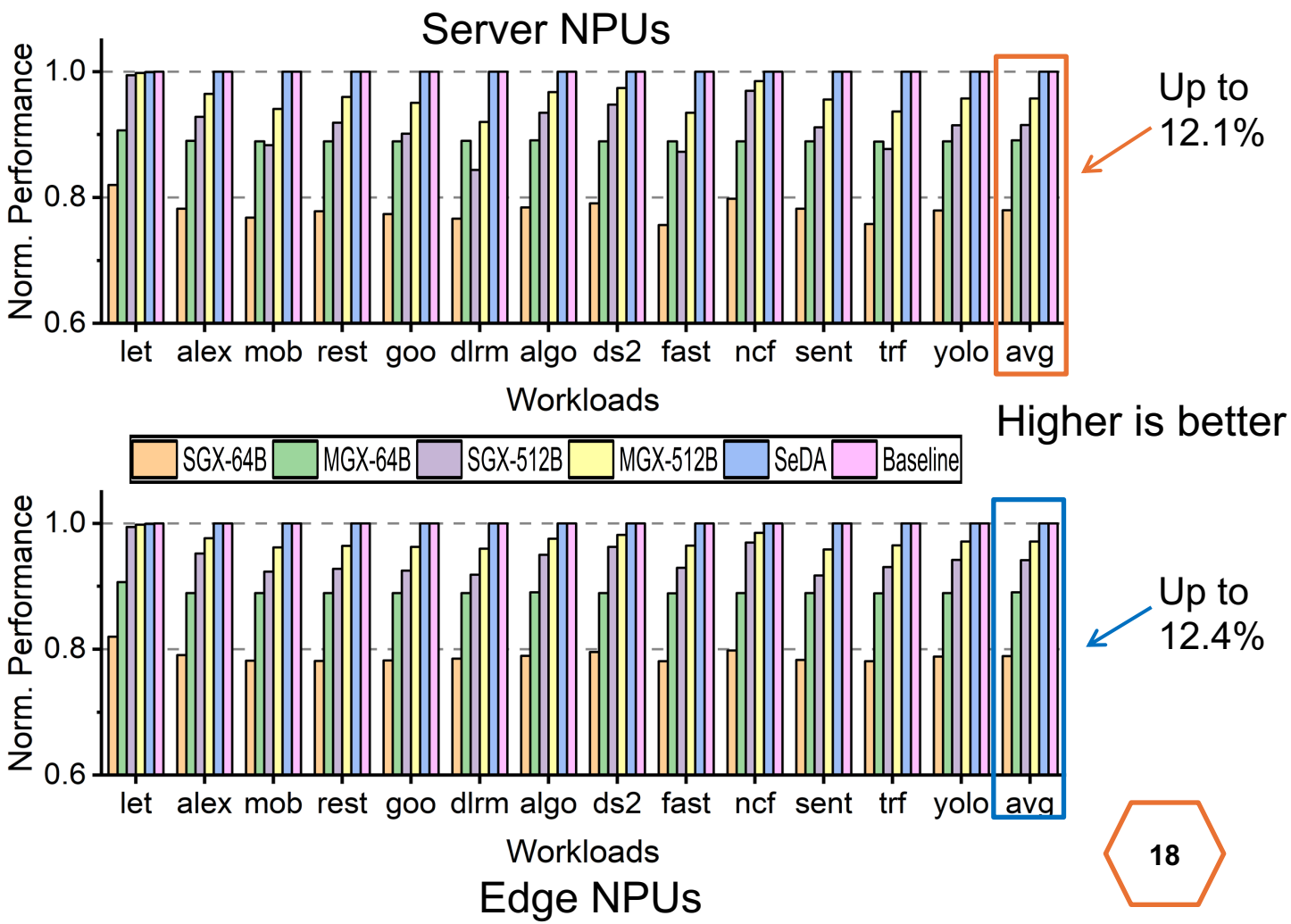
1) SeDA reduces authentication metadata by multi-level integrity verification
2) Off-chip memory traffic reduces over 22% for both server and edge NPUs



# Performance Comparison

- Traffic comparison with five protection schemes:
  - SGX-64B, SGX-512B, MGX-64B, MGX-512B and unsecure baseline
  - 64B and 512B indicate different security granularities

1) SeDA utilizes bandwidth-aware encryption to efficiently generate OTPs and multi-level integrity verification to reduce memory access
2) Performance improve over 12% for both server and edge NPUs



# Summary

- **Problem**
  - Secure DNN accelerator increases off-chip memory access and degrades the performance
- **Solution**
  - Bandwidth-aware encryption reduces hardware resources for confidentiality protection
  - Multi-level integrity verification reduces off-chip memory access of security metadata
- **Evaluation**
  - Reduce the performance overhead by over 12% compared to of state-of-the-art approaches

# Thank You!

# Q & A



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